

Take a sequence $\{a_n\}$, we can add all the terms together

$$a_1 + a_2 + a_3 + \dots$$

an infinite number of terms, this is denoted

$$\sum_{i=1}^{\infty} a_i, \text{ called a series}$$

but does this always make sense? Certainly many series diverge, take $\sum_{i=1}^{\infty} n = 1 + 2 + 3 + \dots$.

But we will see that some will converge.

Consider, again, Xeno's Paradox.

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

we notice that

$$\sum_{n=1}^1 \frac{1}{2^n} = \frac{1}{2}, \quad \sum_{n=1}^2 \frac{1}{2^n} = \frac{3}{4}, \quad \frac{7}{8}, \frac{15}{16}, \dots$$

in fact,

$$\sum_{i=1}^n \frac{1}{2^i} = \frac{2^n - 1}{2^n}, \text{ and } \lim_{n \rightarrow \infty} \frac{2^n - 1}{2^n} = 1$$

So $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$, which is expected.

Like with Sequences, we discuss Convergent and divergent Series. To do so, we make a new sequence

Take $\sum a_n$, consider the partial sums

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$\vdots$

$$S_n = \sum_{i=1}^n a_i$$

Then look at the sequence $\{S_n\}$ to decide if it converges or diverges

This, at least, the theory. In practice there are simpler methods.

Def:

Take a series $\sum a_n$ where S_n is the n^{th} partial sum.

The sequence $\{S_n\}$ is convergent if $\lim_{n \rightarrow \infty} S_n = S$ exists

in which case

$$\sum_{n=1}^{\infty} a_n = S$$

and S is the sum of the series.

If $\{S_n\}$ is divergent, so too is $\sum a_n$

Ex:

Suppose $\sum_{n=1}^{\infty} a_n$ has partial sum $S_n = \frac{2n}{3n+5}$

$$\text{Then } \lim_{n \rightarrow \infty} \frac{2n}{3n+5} = \frac{2}{3} \text{ so } \sum_{n=1}^{\infty} a_n = \frac{2}{3}$$

Ex: Telescoping Series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}, \quad \text{notice } \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$\text{So } \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

↳ Note, we cannot write $\sum \frac{1}{n} - \sum \frac{1}{n+1}$

Then

$$S_n = \sum_{i=1}^n \left(\frac{1}{i} - \frac{1}{i+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$= 1 - \frac{1}{n+1}$$

$$S_\infty = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 1 - \frac{1}{n+1} = 1$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

Geometric Series

One of the more important type of series are called Geometric Series. They take the form

$$a + ar + ar^2 + ar^3 + \dots = \sum_{n=1}^{\infty} ar^{n-1} \quad a \neq 0$$

If $r = 1$,

$$S_n = a + a + \dots = na \quad \text{which diverges}$$

If $r \neq 1$

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^n$$

$$rS_n + a = a + ar + ar^2 + \dots + ar^{n-1} + ar^n$$

$$rS_n + a = S_n + ar^n$$

$$a - ar^n = S_n - rS_n$$

$$S_n = \frac{a - ar^n}{1 - r}$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

If $|r| > 1$, then $\lim_{n \rightarrow \infty} r^n = \infty$ and $\{S_n\}$ diverges

If $|r| < 1$, $\lim_{n \rightarrow \infty} r^n = 0$ and

$$\lim_{n \rightarrow \infty} S_n = \frac{a}{1 - r}$$

Thm:

The geometric Series $\sum_{n=1}^{\infty} ar^{n-1}$ Converges if $|r| < 1$, in which case

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1 - r}$$

Note, $\frac{a}{1-r}$ is just $\frac{\text{first term}}{1 - \text{Common ratio}}$

Ex:

Find the following sum

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \dots$$

Common ratio is $-\frac{2}{3}$, $|-\frac{2}{3}| < 1$
first term is 5

$$\text{Sum is } \frac{5}{1 - (-\frac{2}{3})} = \frac{5}{\frac{5}{3}} = \boxed{3}$$

Ex

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$$

$$\begin{aligned} &= \sum_{n=1}^{\infty} 4^n 3^{-(n-1)} = \sum_{n=1}^{\infty} 4 \cdot 4^{n-1} \left(\frac{1}{3}\right)^{n-1} \\ &= \sum_{n=1}^{\infty} 4 \left(\frac{4}{3}\right)^{n-1} \end{aligned}$$

$r = \frac{4}{3} > 1$, So Series diverges

Ex: write $2.\overline{317}$ as a ratio of integers

$$2.\overline{317} = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \dots$$

$$= \frac{23}{10} + \sum_{n=1}^{\infty} \frac{17}{10^3} \left(\frac{1}{100}\right)^{n-1}$$

$$= \frac{23}{10} + \frac{\frac{17}{10^3}}{1 - \frac{1}{100}}$$

$$= \frac{23}{10} + \frac{17}{10^3} \cdot \frac{100}{99}$$

$$= \frac{23}{10} + \frac{17}{990}$$

$$= \frac{1147}{995}$$

Ex: $0.\overline{9} = 1$

$$0.\overline{9} = \frac{9}{10} + \frac{9}{100} + \frac{9}{10^3} + \dots$$

$$\begin{aligned} &= \sum_{n=1}^{\infty} 9 \left(\frac{1}{10}\right)^n = \frac{9/10}{1 - 1/10} \\ &= \frac{9/10}{9/10} \\ &= 1 \end{aligned}$$

Ex: let $|x| < 1$

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$$

$$= \frac{1}{1-x}$$

Divergence Test

Remember, a series is divergent if the sequence of partial sums $\{S_n\}$ diverges.

Ex: (Harmonic Series)

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots \quad \text{is divergent}$$

Pf/

$$S_2 = 1 + \frac{1}{2}$$

$$S_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + 1$$

$$S_8 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > 1 + \frac{1}{2} + \left(\frac{1}{3}\right) + \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right) \\ = 1 + \frac{3}{2}$$

$$S_{16} = 1 + \dots + \frac{1}{16} > 1 + \frac{4}{2}$$

$$S_{32} > 1 + \frac{5}{2}, S_{64} > 1 + \frac{6}{2}, S_{2^n} > 1 + \frac{n}{2}$$

$$\lim_{n \rightarrow \infty} S_{2^n} = \lim_{n \rightarrow \infty} 1 + \frac{n}{2} = \infty$$

$$S_c \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}$$

Thm:

If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$

$$\text{Pf/ } S_n = a_1 + a_2 + \dots + a_n, \quad \lim_{n \rightarrow \infty} S_n = L$$

$$S_o \quad a_n = S_{n+1} - S_n$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} S_{n+1} - S_n = L - L = 0$$

Thm (Divergence Test)

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges

Ex:

$$\sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4}$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{5n^2 + 4} = \frac{1}{5} \quad S_c \sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4} \text{ diverges}$$

Properties of Convergent Series

Thm:

If $\sum a_n$ and $\sum b_n$ are convergent,
so too is $\sum c a_n, \sum (a_n \pm b_n)$

Moreover,

$$\cdot \sum c a_n = c \sum a_n$$

$$\cdot \sum (a_n \pm b_n) = \sum a_n \pm \sum b_n$$

Ex:

$$\sum_{n=1}^{\infty} \frac{3}{n(n+1)} + \frac{1}{2^n}$$

$$\swarrow$$
$$3 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 3(1) = 3$$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1/2}{1 - 1/2} = 1$$

$$\sum \frac{3}{n(n+1)} + \frac{1}{2^n} = 3 + 1 = \boxed{4}$$

Note: finite numbers don't affect convergence or divergence. So, if

$\sum a_n$ converges

$\sum b_n$ diverges

$$\sum (a_n \pm b_n) = \sum a_n \pm \sum b_n \text{ diverges}$$

The same is not true if both a_n and b_n diverge